Multi-Step Proportion Problems

Teacher's Edition

The Period

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Must Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

Must not A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

Target I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

MTR **MTR.1.1** — Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going · **MTR.6.1** — Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer · **MTR.7.1** — boards 8–9: a plan's area in square meters and a choice of car

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A multi-step problem is one unit rate used again — name what each step finds.
2. Same unit first; then compare.
3. A recipe's k comes from the yields; then every ingredient, then the pantry.

Slide by Slide

1. **Multi-Step Proportion Problems.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the unit rate every board today starts from. *1 retrieves 5.08–09 — × the ratio whose 1 is kilograms (yesterday). 2 retrieves 5.04 — drawing × rate (earlier this unit). 3 retrieves 4.04 — a two-step with decimals (last unit). 4 retrieves MA.6.AR.3.2 — a unit rate; today every board starts from one (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.816 lb 2. 150 km 3. 9 4. 60 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** A car gets 23 miles per gallon and its tank holds 17.5 gal: 23 × 17.5 = 402.5 mi on a tank.
6. **Notes.** 100 km/h against 65 mi/h: nothing compares until the units match.
7. **Notes.** 12 cookies take 1½ cups of flour.
8. **Example 1 — Two rates in different units.** Ask which unit the comparison will be made in before anyone computes.
9. **Example 1 — Worked.** A: 8.05 km in ⅔ h is 12.075 km/h; B: 9 km in ¾ h is 12 km/h.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A: 6.44 km in ½ h is 12.88 km/h; B: 7 km in 7⁄12 h is 12 km/h.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A recipe scaled, then checked.** Ask what k is before anyone touches the rice.
13. **Example 2 — Worked.** k = 2.5, so 6¼ cups; 5 cups is 2 recipes, 16 servings.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 2, 3 cups; 4½ cups is 3 recipes, 18 servings.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** A car gets 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank? → ***420 mi*** *1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added*
17. **18–19. Board 2.** A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take? → ***40 min*** *49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value*
18. **20–21. Board 3.** Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn? → ***$435*** *The 2 mi is not needed. 87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice*
19. **22–23. Board 4.** A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) → ***D — 3 tanks*** *(A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [7.AR.4.5: the steps not named — Notes I] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg? → ***120 g*** *12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100*

26–27. Board 6. $1\,\text{mi} = 1.61\,\text{km}$. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster? → ***C — cyclist B*** *(A) 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [7.AR.4.5: a rate is per unit of time — Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. Paint is mixed white : blue : grey = $2:4:5$. A batch uses 8 gal of white. How many gallons is the whole batch? → ***44 gal*** *20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]; 32 — 8 × k, the white scaled again*

30–31. Board 8. A plan is drawn at $1\,\text{in} : 4\,\text{ft}$; a room is 2 in by 2.5 in on it. $1\,\text{ft} = 0.305\,\text{m}$. Find the room's area in square meters. → ***About 7.44 m²*** *24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6]*

32–33. Board 9. $1\,\text{mi} = 1.61\,\text{km}$. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain. → ***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.*** *'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long'*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–3 use one rate once, backwards, and on a total, the numbers growing; 4 asks the second step as a vote; 5 changes the unit inside the rate. The pattern breaks at 6, where the comparison cannot be made until the units match. 7 is part-to-whole; 8 and 9 are the test's transfer shapes rehearsed. The independent set walks the same path: a range, the tanks, a recipe, a comparison in one unit, a mixture, a plan's area in square meters.

Misconceptions to Watch

Board 1. 1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]

Board 2. 49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]

Board 3. 87 — one day's pay [Notes I: the question asks for five days]

Board 4. 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]

Board 5. 12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]

Board 6. 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]

Board 7. 20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]

Board 8. 24.4 — 80 ft^2 × 0.305, the ratio once on an area [5.08–09 Notes II]

Board 9. 'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.